Appendix 5. Within-study ablation results

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Each row compares two versions of the same model evaluated on the same DS2 test set. The versions differ in a single component, except for Tao Y [40], where several modules were added together. These values correspond to Figure 4 of the article. V-class values are given when the study reported them.

Study	Comparison (before, then after)	S Se/PPV before	S Se/PPV after	V Se/PPV before	V Se/PPV after
RR information					
Zhang J 2021 [24]	Without RR ratios, then with RR ratios	6.3/42.2	78.8/90.8	88.7/91.3	92.5/94.3
Ran 2024 [34]	Standard focal loss (not the final loss) without RR features, then with RR features*	4.3/4.8	84.2/37.8	80.6/90.9	94.1/66.3
Li Z 2025 [28]	Adaptive focal loss without RR ratio, then with RR ratio	22.9/18.2	72.1/90.3	NR	NR
Li Z 2025 [28]	Raw previous RR, then RR ratio	69.8/64.0	72.1/90.3	NR	NR
Niu 2020 [50]	CNN alone, then CNN with six RR features*	63.5/35.7	81.9/59.1	NR	NR
Class rebalancing					
Wang T 2020 [30]	Cross-entropy, then focal loss	80.7/57.1	81.4/68.3	93.5/84.0	93.7/91.1
Zubair 2024 [27]	Imbalanced data, then translation-based oversampling	93.9/85.4	96.3/95.3	90.1/96.3	97.3/97.9
Xu P 2022 [31]	Without, then with minority augmentation	75.9/71.6	83.1/82.5	89.9/83.0	93.5/90.0
Midani 2024 [36]	Without, then with SMOTE (5 classes)	48.3/67.0	93.0/96.0	100.0/97.6	100.0/99.3
Lyu 2023 [38]	Without, then with SMOTE (5 classes) *	99.7/99.6	99.2/95.9	NR	NR
Input context and false-positive control					
Essa 2021 [37]	One beat, then five beats as input*	67.2/17.4	81.6/18.4	NR	NR
Essa 2021 [37]	Ensemble alone, then with verification network	80.1/20.6	65.5/66.2	95.6/91.4	93.9/94.5
Wang Y 2023 [35]	Without, then with post-hoc RR rule	88.9/26.0	73.9/75.6	NR	NR
Tao Y 2026 [40]	Plain transformer, then full model	56.0/86.1	75.5/91.2	NR	NR
Adaptation and training data					
Chen M 2020 [47]	Fixed model, then adapted model	64.4/66.4	78.6/88.7	90.7/85.5	95.8/93.9
Wang G 2021 [49]	Fixed model, then adapted model	71.9/82.9	76.5/90.2	92.0/80.2	94.0/92.3
Lyu 2023 [38]	All of DS1, then half of DS1 for training	99.7/99.6	37.8/75.2	100.0/99.9	99.9/98.9

Values in %. Se, sensitivity. PPV, positive predictive value. NR, not reported for that comparison. Reference numbers follow the main article. * “After” values may differ from Table 1 when the ablation used a variant of the final model.